

# Reactive Extrusion Of Poly(ethylene terephthalate)–(Ethylene/Methyl Acrylate/Glycidyl Methacrylate)–Organoclay Nanocomposites

Elif Alyamac, Ulku Yilmazer

Department of Chemical Engineering, Middle East Technical University, Ankara 06531, Turkey

This study was conducted to investigate the effects of component concentrations and addition order of the components on the final properties of ternary nanocomposites composed of poly(ethylene terephthalate), organoclay, and an ethylene–methyl acrylate–glycidyl methacrylate (E-MA-GMA) terpolymer acting as an impact modifier for PET. In this context, first, the optimum amount of the impact modifier was determined by melt compounding binary PET-terpolymer blends in a corotating twin-screw extruder. The amount of the impact modifier (5 wt%) resulting in the highest Young's modulus and moderate elongation at break was selected owing to its balanced mechanical properties. Thereafter, by using 5 wt% terpolymer content, the effects of organically modified clay concentration and addition order of the components on the properties of ternary nanocomposites were systematically investigated. Mechanical testing revealed that different addition orders of the materials significantly affected the mechanical properties. Among the investigated addition orders, the best sequence of component addition (PI-C) was the one in which poly(ethylene terephthalate) was first compounded with E-MA-GMA. Later, this mixture was compounded with the organoclay in the subsequent run. In X-ray diffraction analysis, extensive layer separation associated with delamination of the original clay structure occurred in PI-C and CI-P (Clay + Impact Modifier followed by PET) sequences with both 1 and 3 wt% clay contents. X-ray diffraction patterns showed that at these conditions exfoliated structures resulted as indicated by the disappearance of any peaks due to the diffraction within the consecutive clay layers. *POLYM. COMPOS.*, 28: 251–258, 2007. © 2007 Society of Plastics Engineers

## INTRODUCTION

In recent years, it is found that when fillers are dispersed in polymers on the nanometer scale, composites obtained show improved properties compared with conventional polymer composites. Among the properties improved by the

presence of nanofillers are mechanical, thermal, physical, and barrier properties. This new class of composites is called polymer matrix nanocomposites. Today, studies are being conducted globally, using almost all types of polymer matrices. Consequently, current reports on poly(ethylene terephthalate)-based nanocomposites [1–5] and impact modification of poly(ethylene terephthalate) [6] exist in the literature.

Nanocomposites with impact modified polyamide matrices showing improved toughness have been investigated [7, 8]. However, there are no studies on nanocomposites formed from organically modified clay as the reinforcing agent and impact modified poly(ethylene terephthalate) as the matrix. Poly(ethylene terephthalate) (PET) is a thermoplastic polyester having poor impact resistance and high notch sensitivity. Additionally, in PET/clay nanocomposites, addition of clay sometimes imparts drawbacks to the resulting material, such as brittleness. For this reason, impact modification of PET can be achieved by dispersing elastomeric polymers in the polymer matrix.

In this study, the effects of component concentrations and addition order of the components on the final properties of ternary nanocomposites composed of amorphous poly(ethylene terephthalate) matrix, organically modified clay, and an ethylene–methyl acrylate–glycidyl methacrylate (E-MA-GMA) terpolymer were systematically investigated. The morphology was analyzed by X-Ray Diffraction and Scanning Electron Microscopy. Flow characteristics, melting, and crystallization behavior of the compositions were also studied by Melt Flow Index Measurements and Differential Scanning Calorimetry Analysis respectively.

## EXPERIMENTAL

### Materials

Amorphous poly(ethylene terephthalate) with an intrinsic viscosity of 0.57 dl/g was purchased from DupontSA Company, Adana, Turkey. Cloisite 25A, a natural montmo-

Correspondence to: Ulku Yilmazer; e-mail: yilmazer@metu.edu.tr  
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rillonite modified with a quaternary ammonium salt, was purchased from Southern Clay Products, USA. Lotader GMA AX8900, ethylene-methyl acrylate-glycidyl methacrylate (E-MA-GMA) terpolymer was purchased from Atofina Chemicals. Lotader GMA AX8900 resin is an impact modifier for extrusion or blow molding grades of polyesters. The epoxy functionality of GMA can react with the carboxyl and hydroxyl end groups of PET in the melt phase to form a graft copolymer. For this reason, Lotader GMA AX8900 may also be called as a functionalized polymer used as a compatibilizer.

### Preparation of Nanocomposites

A Thermoprism TSE 16 TC twin screw extruder was used for the preparation of all the formulations. The specifications of this extruder are: diameter: 16 mm, co-rotating, fully intermeshing screws with barrel length of 384 mm. The screw speed was 350 rpm and the barrel temperature profile was 200, 270, 270, 270, 270°C. The addition orders investigated can be summarized as follows: P, I, and C stand for PET, impact modifier (E-MA-GMA), and clay respectively. The first two letters indicate the materials mixed in the first run. This mixture was compounded with the third ingredient in the subsequent run. The following sequences were prepared: Sequence 1 (PI-C),

Sequence 2 (PC-I),

Sequence 3 (CI-P),

Sequence 4 (All-S) Simultaneous feeding of all materials.

In addition to the ternary nanocomposites prepared by different addition orders explained above, binary PET/clay nanocomposites designated as PC were also melt blended to be able to compare the effects of impact modifier that was added as the third component in the nanocomposites. All one-step mixing formulations were converted into two-step mixing ones via extruding the product once more. This was done to make the results comparable with each other. After melt compounding, a laboratory scale injection molding machine (Microinjector, Daca Instruments) was used to mold the specimens of all melt-compounded formulations.

### Characterization

X-ray diffraction patterns were collected with an X-ray diffractometer (Philips, PW-3710). Cu K $\alpha$  ( $\lambda = 1.54 \text{ \AA}$ ) radiation, generated at a voltage of 40 kV and current of 55 mA, was used as an X-ray source. Diffraction angle  $2\theta$  was scanned from 1 to  $10^\circ$  at a scanning rate of  $3^\circ/\text{min}$  and a step size of  $0.02^\circ$ . Scanning electron microscopy (SEM) analysis was performed by a JEOL JSM-6400 low voltage scanning electron microscope. The analysis was conducted on fracture surfaces of impact specimens. Samples that were not fractured by impact loading were fractured under liquid nitrogen. In all cases, the fractured surfaces were gold-coated and mounted on brass stages prior to viewing with the scanning electron microscope. Melt flow index (MFI)

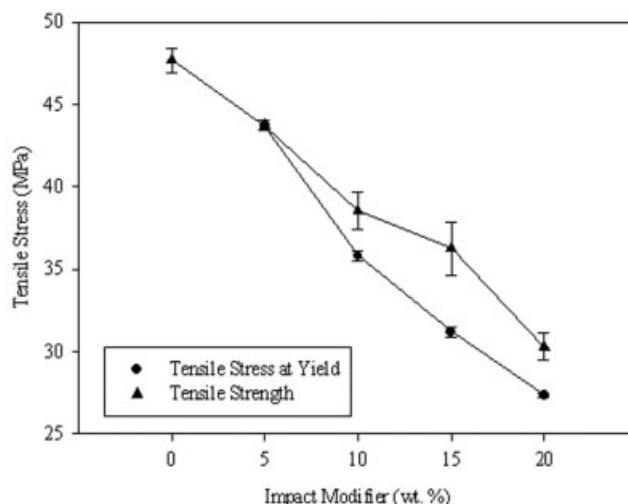


FIG. 1. Tensile stress at yield and tensile strength of PET/impact modifier blends as a function of the impact modifier content.

measurements were carried out according to ASTM D1238–79 using Omega Melt Flow Indexer. Conditions of temperature and load were selected as 260°C and 2.16 kg respectively, which were in accordance with material specifications. Mechanical tests were performed at 23°C. At least seven specimens were tested for each formulation. For each series of tests, the arithmetic mean of all values obtained was calculated and reported as the average value for the particular property in question. Standard deviation was also calculated according to ASTM standards. A Lloyd 30K universal testing machine was used to conduct tensile testing of molded samples. The test was performed conforming to ASTM D638M-91a. Charpy impact tests were carried out by using a Pendulum Impact Tester of Coesfeld Material Test, based on Test Method-I Procedure A in ASTM D256–91a. Differential scanning calorimetry (DSC) measurements were carried out with a General V4.1.C DuPont 2000. In DSC analysis, 10 mg of sample was cut from the center of an injection molded specimen and was heated from room temperature to 280°C at a rate of 20°C/min. The samples were scanned under nitrogen atmosphere.

## RESULTS AND DISCUSSION

### Effect of Impact Modifier

In order to optimize the impact modifier (E-MA-GMA) content for ternary nanocomposites composed of PET, impact modifier, and clay, mechanical behavior of binary PET/E-MA-GMA blends was studied. At the end, the impact modifier content providing balanced mechanical properties was determined.

Figure 1 shows that addition of the impact modifier decreases the tensile strength and the tensile stress at yield. Pure PET does not show yielding; the composition with 5 wt% impact modifier shows yielding but does not undergo

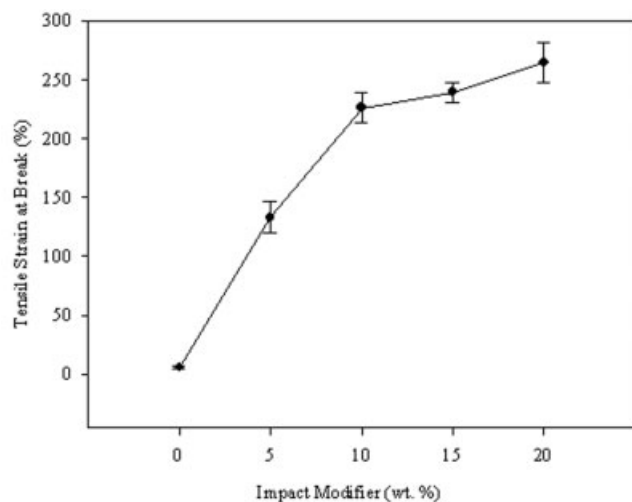


FIG. 2. Tensile strain at break values of PET/impact modifier blends as a function of the impact modifier content.

strain hardening. Blends with 10–20 wt% impact modifier exhibit yielding followed by strain hardening. Percent tensile strain at break values of these blends are shown in Fig. 2. The elongation at break increases significantly, jumping from 5% for pure PET to between 100 and 300% for the modified PET, depending on the modifier concentration. Young's modulus values of PET/impact modifier blends are shown as a function of impact modifier content in Fig. 3. Pure E-MA-GMA has a reported Young's modulus of 8 MPa. Young's modulus decreases from 1,155 MPa for not extruded pure PET to nearly 600 MPa for PET containing 20 wt% impact modifier. Use of 5 wt% modifier leads to a Young's modulus of 910 MPa. On the other hand, PET with 5 wt% impact modifier exhibits lower elongation at break compared with the ones containing 10, 15, or 20 wt% modifier. Thus, in order to have balanced mechanical properties, 5 wt% impact modifier was chosen as the E-MA-

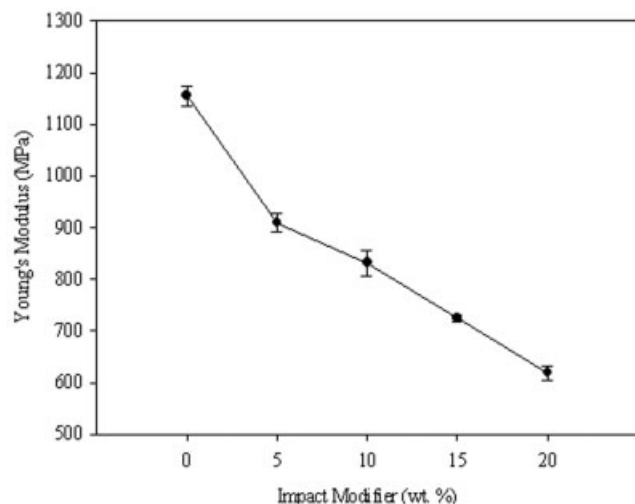


FIG. 3. Young's modulus values of PET/impact modifier blends as a function of the impact modifier content.

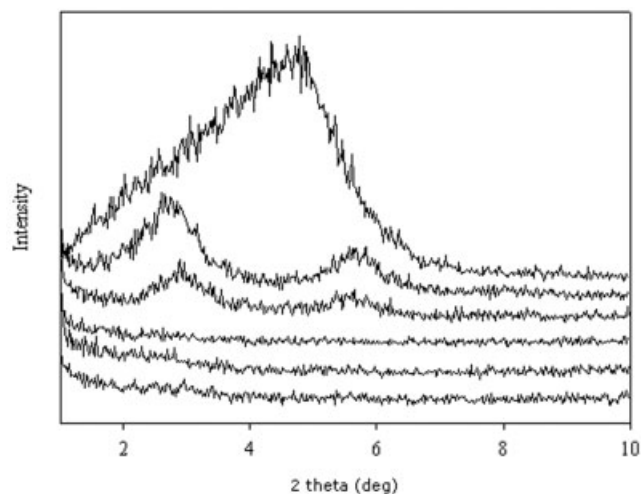


FIG. 4. X-ray diffraction patterns of nanocomposites containing 1 wt% clay (From top to bottom: C25A, PC-I, PC, All-S, CI-P, and PI-C).

GMA content in the ternary nanocomposites prepared subsequently.

#### X-Ray Analysis

Figure 4 shows XRD patterns of nanocomposites with 1 wt% clay (Cloisite 25A) loading. The y-axis is shifted for clarity. The d-spacing indicates the interlayer spacing of the silicate layers, which is calculated from the peak position using Bragg's equation. In this figure, Cloisite 25A has this peak at  $2\theta$  of  $4.92^\circ$ , which corresponds to a d-spacing of  $17.94 \text{ \AA}$ . For PI-C, CI-P, and All-S sequences with 1 wt% clay content, no peak is detected by XRD, which suggests that they have an exfoliated structure.

In Fig. 4, the d-spacing is increased from  $17.94 \text{ \AA}$  for pure Cloisite 25A to  $32.39 \text{ \AA}$  for PC-I containing 1 wt% clay. Whereas, the d-spacing of  $30.44 \text{ \AA}$  for PC containing 1 wt% clay is relatively smaller than that of the nanocomposite having impact modifier as a third component. This indicates that the presence of impact modifier prepared with PC-I mixing sequence has an effect on the intercalation of the silicate layers in the material.

XRD patterns of nanocomposites with 3 wt% clay content are presented in Fig. 5. It is apparent from Fig. 5 that after melt blending, the intensity of the diffraction peak corresponding to C25A is reduced while a set of new peaks appears. In Fig. 5, PC-I, PC, and All-S patterns contain diffraction peaks characteristic of the intercalated structure. Whereas, CI-P and PI-C represent an exfoliated structure since no peak is detected by XRD. This implies that the polymer that enters between the galleries of clay pushes the platelets far enough apart so that the platelets may not be parallel to each other indicating an exfoliated structure. In all the exfoliated nanocomposites, clay layers are subjected to high shear stresses as supported by the melt flow index values.

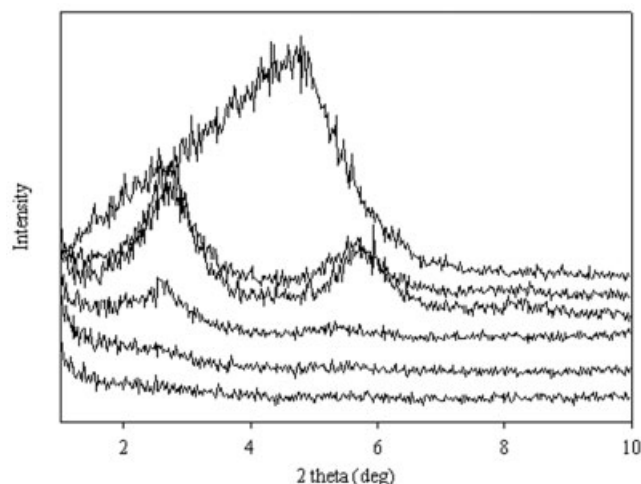


FIG. 5. X-ray diffraction patterns of nanocomposites containing 3 wt% clay (From top to bottom: C25A, PC-I, PC, All-S, CI-P, and PI-C).

### MFI Measurements

In Table 1, melt flow index values of all the formulations are given. The following observations are made:

- The MFI of pure PET increases from 610 to 809 (i.e. the viscosity decreases) upon extrusion indicating chain scission.
- The MFI of E-MA-GMA is much lower than that of PET under the same load and at the same temperature. Thus,

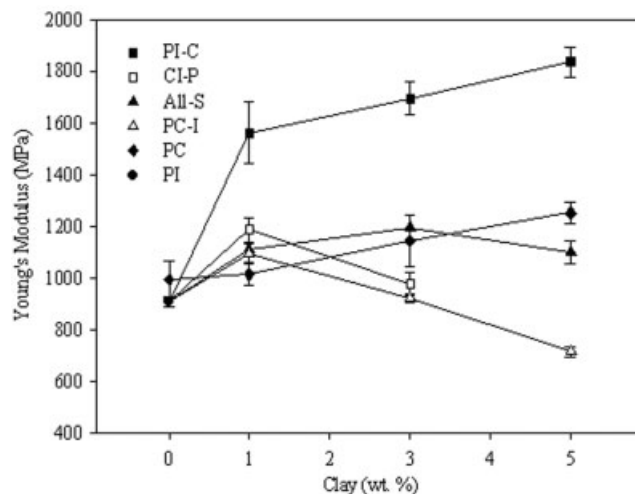


FIG. 6. Young's modulus values of all formulations as a function of clay concentration.

TABLE 1. MFI values of all formulations.

| Polymer                 | MFI (g/10 min) |
|-------------------------|----------------|
| Pure PET (not extruded) | 610            |
| E-MA-GMA                | 26             |
| PI, E-MA-GMA (wt %)     |                |
| 5                       | 166            |
| 10                      | 147            |
| 15                      | 143            |
| 20                      | 61             |
| PC, Clay (wt %)         |                |
| 0                       | 809            |
| 1                       | 496            |
| 3                       | 358            |
| 5                       | 262            |
| CI-P, Clay (wt %)       |                |
| 1                       | 140            |
| 3                       | 124            |
| PC-I, Clay (wt %)       |                |
| 1                       | 236            |
| 3                       | 207            |
| 5                       | 184            |
| PI-C, Clay (wt %)       |                |
| 1                       | 115            |
| 3                       | 112            |
| 5                       | 110            |
| All-S, Clay (wt %)      |                |
| 1                       | 214            |
| 3                       | 183            |
| 5                       | 167            |

in PI blends; as the impact modifier content increases, the MFI decreases.

- Clay decreases the MFI (increases the viscosity) since it acts as a filler, as observed from PC materials.
- Upon addition of E-MA-GMA to PC blends, the MFI decreases (the viscosity increases) as expected. However, this decrease in MFI is not the same for all the mixing sequences. CI-P and PI-C give the lowest MFI (highest viscosity). It should be noted that these sequences gave rise to exfoliated structures.

In the case of PI-C, the epoxy functionality of GMA reacts with PET [9] and forms a viscous mixture in the first extrusion step as supported by the decrease in MFI. Thus, in the second extrusion step, high shear stresses are applied on the clay layers and exfoliation takes place, since the shear stress increases with increasing viscosity. Likewise, in the case of CI-P, E-MA-GMA also exfoliates the clay layers owing to its high viscosity. However, in PC-I, this mechanism does not take place. The pure PET cannot exfoliate the clay layers, owing to its low viscosity, whereas the PI-C shows the highest viscosity among the ternary systems studied.

### Effects of Addition Order and Clay Concentration on the Mechanical Behavior

Figure 6 displays Young's modulus values of both binary and ternary nanocomposites with respect to the amount of clay. Young's modulus is greatly influenced by the content of clay and by the addition sequence of the three components. The characteristic behavior of polymer/layered silicate nanocomposite materials is observed [10]. Obviously, Young's modulus of PET/organoclay nanocomposites (PC) increases with increasing clay content since any filler whose modulus is greater than that of the polymer increases the modulus of a mixture with that polymer. For PI-C, there is

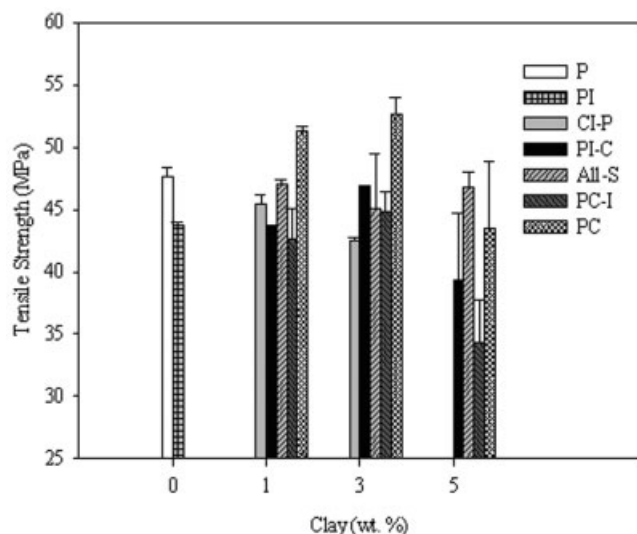


FIG. 7. Tensile strength as a function of clay content.

a sharp increase of Young's modulus for very small clay loadings (1 wt%) followed by a much slower increase beyond 1 wt% loading. On the other hand, with increasing clay content, Young's modulus values of All-S, CI-P, and PC-I do not change markedly up to loading levels of 3 wt% clay when compared with Young's modulus of PI (0 wt% clay). These observations support the fact that in the ternary systems better dispersion of the clay in PET and strong interaction between PET and the clay are essential for achieving high Young's modulus. While the highest Young's modulus is revealed at 5 wt% clay for PI-C, the lowest value obtained is at 5 wt% clay loading for PC-I. The reactive extrusion of the PET and the impact modifier prior to compounding with the clay must have resulted in promoting the exfoliation of clay layers owing to the high molecular weight and the high viscosity obtained in comparison to pure PET.

Figures 7 and 8 exhibit the tensile strength and the tensile strain at break of the materials studied. It is observed that addition of E-MA-GMA decreases the tensile strength of pure PET and PET-clay binary nanocomposite materials, since E-MA-GMA has a lower tensile strength than does PET. The order of the addition does not significantly affect the tensile strength of the nanocomposites. In general, the tensile strength values of the materials with 1 and 3 wt% clay contents are higher than those of the materials with 5 wt% clay content. This is probably due to higher levels of exfoliation at low clay contents as observed by XRD.

On the other hand, the tensile strain at break values shown in Fig. 8 are significantly higher at 1 wt% clay content owing to high levels of exfoliation. While pure PET does not show yielding and breaks at very low strains, addition of E-MA-GMA results in a ductile material that elongates to very high extension after the yield point. At 1 wt% clay content, the mixing sequences of CI-P, PI-C, and, to a degree, All-S exhibit yielding, followed by elongation to very high strains. As observed earlier by XRD and MFI,

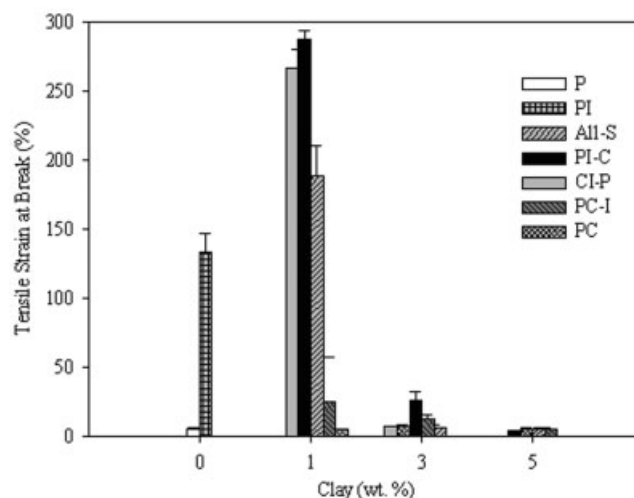


FIG. 8. Tensile strain at break as a function of clay content.

these mixing orders produce exfoliated structures. At higher clay content (3 wt%), materials mixed with these sequences still exhibit yield, but do not elongate to high strains owing to the stress concentration effect of unexfoliated clay particles. The strain at break further decreases at 5 wt% clay concentration.

Figure 9 confirms the previous discussion. At 0 wt% clay content, pure PET does not exhibit yielding whereas PET/E-MA-GMA blend does. At 1 wt% clay content, PI-C, CI-P, All-S, and PC-I mixing sequences show yielding. At 3 wt% clay content, PI-C, CI-P, and All-S mixing sequences also exhibit yielding. The level of yield stress, when yielding exists, is insensitive to the mixing order and the clay content. However, at 5 wt% clay content, the nanocomposites break without yielding.

The unnotched Charpy Impact strength of all the materials at room temperature is summarized in Fig. 10. The pristine PET has a poor impact resistance with a value of

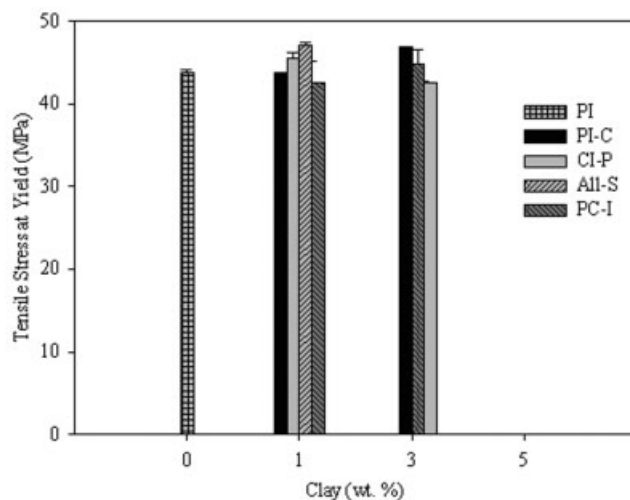


FIG. 9. Tensile stress at yield as a function of clay content.

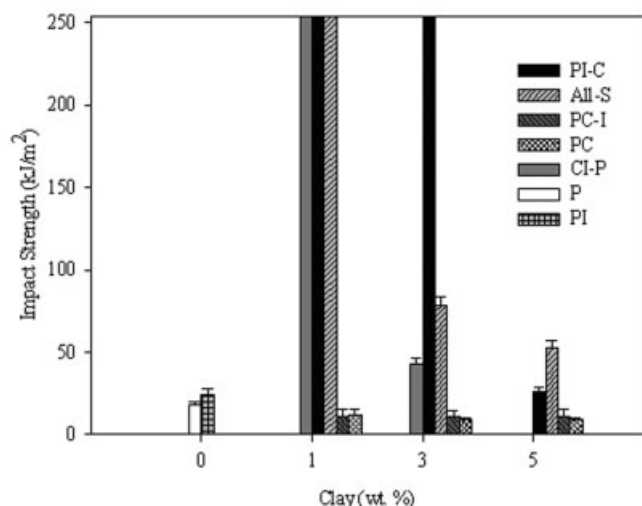


FIG. 10. Impact strength as a function of clay content.

nearly 20 kJ/m<sup>2</sup>. Adding 5 wt% impact modifier did not significantly change the impact strength of pure PET, whereas the impact strength results drastically changed in some mixing sequences and clay contents when the modifier was added as a third component in the PET/clay nanocomposites.

It should be noted that while pure PET specimens always broke completely upon impact, the tough nanocomposites indicated in Fig. 10 did not break, and their impact strength recorded is in excess of 254 kJ/m<sup>2</sup>. This is the maximum value that can be measured with these dimensions of the samples. The impact values are consistent with the presence or absence of yielding. At 0 wt% clay content, the impact strength of the PET increases with the addition of E-MA-GMA. At 1 wt% clay content, the exfoliated structures of CI-P, PI-C, and All-S mixing sequences did not break and thus their impact strength is higher than 254 kJ/m<sup>2</sup>. At 3 wt% clay content, the same is true for the PI-C mixing order. The impact strength values of the materials with CI-P and All-S mixing sequences are still high. The impact strength decreases at 5 wt% clay content, owing to the stress concentration effect of unexfoliated clay.

Based on the impact and the strain at break results, the best sequence of component addition is once more observed as (PI-C). In this case, delaminated clay layers are obtained at 1 and 3 wt% clay contents resulting in best mechanical properties. However, in other addition sequences such as PC-I, the matrix cannot apply high shear stresses owing to its lower viscosity, thus delamination of clay layers does not take place.

#### SEM Analysis

Figure 11 represents the impact fractured surfaces of twice extruded PET at magnification of 3,500 $\times$ . PET has a smooth structure. Almost no fragmentation is observed and few straight crack lines are apparently seen indicating low

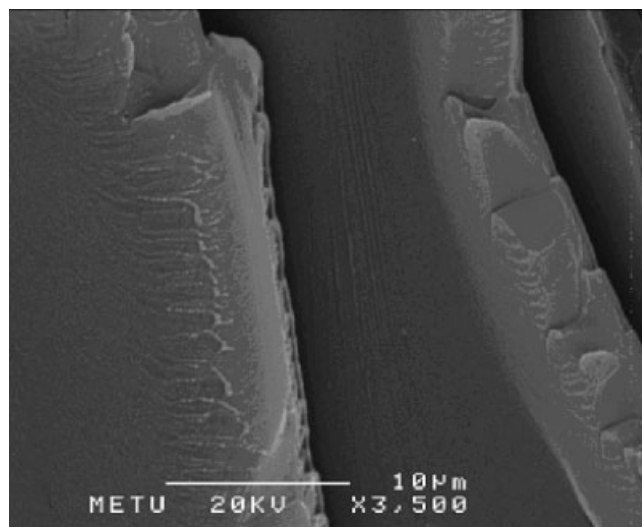


FIG. 11. SEM micrograph of double extruded, pure PET at a magnification of  $\times 3,500$ .

impact strength. In Fig. 12, SEM micrograph of PET/E-MA-GMA blend with 5 wt% E-MA-GMA is shown at a magnification of 3,500 $\times$ . It is obvious that featureless structure of pure PET disappears when melt blended with E-MA-GMA. The continuous and interpenetrated phases of PET and E-MA-GMA are clearly seen suggesting that there is very good adhesion between PET and E-MA-GMA owing to the intermolecular reactions between the two polymers. Figure 13 represents the SEM micrograph of PI-C nanocomposites with 3 wt% clay content. The crack propagation path is very tortuous, since this material is one of the toughest materials as observed by the impact data.

#### Thermal Analysis

As is seen in Table 2, the melting temperature ( $T_m$ ) and glass transition temperature ( $T_g$ ) of all the formulations

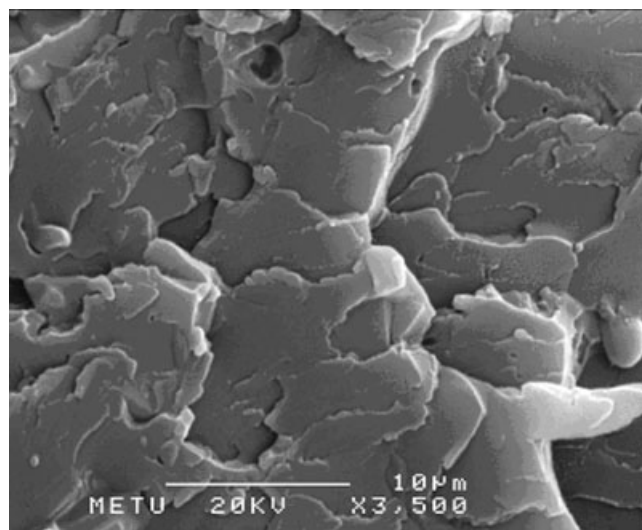


FIG. 12. SEM micrograph of PET/impact modifier blend with 5 wt% impact modifier content at a magnification of  $\times 3,500$ .

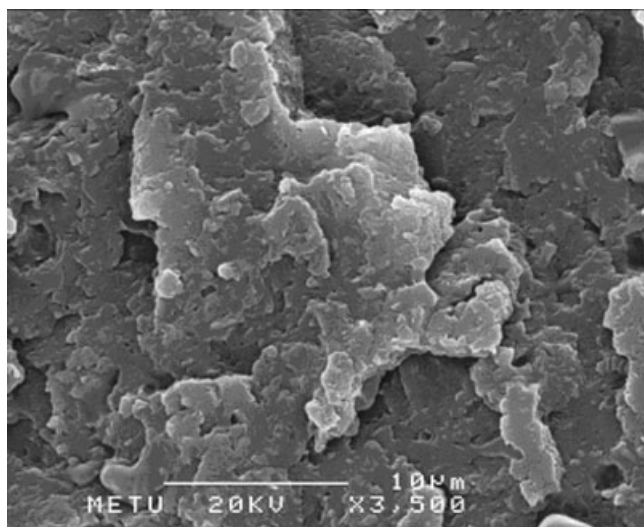


FIG. 13. SEM micrograph of PET/impact modifier/clay nanocomposites for PI-C with 3 wt% clay content at a magnification of  $\times 3,500$ .

remain almost the same; the variations are about 2–4°C. This suggests that the incorporation of both clay and impact modifier do not significantly influence the melting and glass transition behavior of the compositions. However, the highest  $T_g$  values are observed for PI-C with 1 and 3 wt% clay and All-S with 1 wt% clay content, which are 79.4, 79.7, and 79.5°C respectively. It is obvious that relative to the pure PET with a  $T_g$  value of 76.7°C, the composites con-

taining exfoliated clay show higher  $T_g$ . This finding suggests that PET segments confined within the silicate galleries of the clay tend to retard the segmental motion of the PET matrix [11]. As for the crystallization temperature ( $T_c$ ); all compositions show a crystallization peak during heating, and the crystallization temperature decreases by the addition of the clay and impact modifier. For example, the  $T_c$  values of PI-C with 1, 3, and 5 wt% clay contents are 128.5, 122.4, and 118.7°C respectively. The influence of addition order on crystallization temperature is not significant as compared with the effect of clay content. The results on PC and other compositions with clay indicate that the clay has a strong nucleation effect and increases the crystallization rate [12].

## CONCLUSIONS

MFI measurements showed that the MFI of pure PET increased upon reactive extrusion. The MFI of E-MA-GMA is much lower than that of PET under the same load and at the same temperature. Thus, in PI blends, as the impact modifier content increased, the MFI decreased. In addition, clay decreased the MFI of the compositions. Upon addition of E-MA-GMA to PC blends, the MFI decreased. However, this decrease in MFI was not the same for all the mixing sequences. CI-P and PI-C gave the lowest MFI (highest viscosity). In the case of CI-P, the impact modifier delaminated the clay layers owing to its high viscosity. In the case of PI-C, the impact modifier reacted with the PET and increased the viscosity of the mixture; thus again delaminated structures were obtained as supported by various properties observed.

In X-ray analysis, for PI-C, CI-P, and All-S sequences with 1 wt% clay content, no peak was detected by XRD, which suggests that they have an exfoliated structure. X-ray patterns showed that, as the level of exfoliation increased, the peaks decreased in height and got broader.

As for the effect of clay concentration on the mechanical properties of the materials, Young's modulus of PET/clay nanocomposites increased with increasing clay content. However, Young's modulus was greatly influenced by the addition order of PI-C; sharp increase of Young's modulus was observed for very small clay loadings (1 wt%).

The tensile strain at break values, on the other hand, were significantly higher at 1 wt% clay content. While pure PET did not show yielding and fractured at very low strain values, addition of E-MA-GMA resulted in a ductile material that extended to a very high strain after the yield point. At 1 wt% clay content, the mixing sequences of CI-P, PI-C, and to a degree All-S exhibited yielding followed by elongation to very high strains.

The impact strength of tough nanocomposites was recorded in excess of 254 kJ/m<sup>2</sup>. The impact values were consistent with the presence or absence of yielding. At 0 wt% clay content, the impact strength of the PET increased with the addition of E-MA-GMA. At 1 wt% clay content, the exfoliated structures obtained with CI-P, PI-C, and All-S mixing sequences did not break and thus, their impact

TABLE 2. Thermal properties of all formulations.

| Polymer                 | $T_g$ (°C) | $T_c$ (°C) | $T_m$ (°C) |
|-------------------------|------------|------------|------------|
| Pure PET (not extruded) | 76.7       | 138.7      | 253.9      |
| E-MA-GMA                | <(-25)     | —          | 68.8       |
| PI, E-MA-GMA (wt. %)    |            |            |            |
| 5                       | 76.5       | 126.3      | 255.5      |
| 10                      | 76.3       | 130.6      | 255.1      |
| 15                      | 77.7       | 129.0      | 254.3      |
| 20                      | 78.8       | 126.2      | 254.3      |
| PC, Clay (wt. %)        |            |            |            |
| 0                       | 77.5       | 133.8      | 255.6      |
| 1                       | 78.8       | 120.6      | 256.3      |
| 3                       | 78.5       | 121.9      | 256.1      |
| 5                       | 76.9       | 116.1      | 253.8      |
| CI-P, Clay (wt. %)      |            |            |            |
| 1                       | 74.6       | 130.4      | 254.9      |
| 3                       | 78.1       | 121.0      | 253.9      |
| PC-I, Clay (wt. %)      |            |            |            |
| 1                       | 77.8       | 126.0      | 257.3      |
| 3                       | 74.6       | 119.4      | 253.9      |
| 5                       | 76.9       | 116.4      | 255.5      |
| PI-C, Clay (wt. %)      |            |            |            |
| 1                       | 79.4       | 128.5      | 253.8      |
| 3                       | 79.7       | 122.4      | 255.1      |
| 5                       | 73.6       | 118.7      | 253.9      |
| All-S, Clay (wt. %)     |            |            |            |
| 1                       | 79.5       | 126.6      | 257.4      |
| 3                       | 74.8       | 123.6      | 254.8      |
| 5                       | 78.3       | 122.1      | 256.0      |

strength was in excess of 254 kJ/m<sup>2</sup>. At 3 wt% clay content, the same was true for the PI-C mixing order. The impact strength of the materials with CI-P and All-S mixing sequences was still high. However, the impact strength decreased significantly at 5 wt% clay content, owing to the stress concentration effect of clay agglomerates.

DSC analysis showed that the highest  $T_g$  values were observed for PI-C and All-S with 1 wt% clay concentration, which suggested that relative to pure PET, the nanocomposites containing exfoliated clay showed higher  $T_g$ . The crystallization temperature decreased by the addition of the clay, since clay acted as a nucleating agent and decreased the crystallization temperature of pure PET.

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